

Chapter 1

INTRODUCTION

The need for efficient and sustainable solutions in landscaping and agricultural tasks has prompted the development of automated systems. One such innovation is the **solar-powered robotic grass cutter**, which combines renewable energy with robotics to provide an eco-friendly and autonomous solution for lawn maintenance. Traditional grass cutters typically rely on gasoline or electricity, which can contribute to environmental pollution and high operational costs. In contrast, a solar-powered system offers a greener alternative by harnessing the sun's energy, reducing carbon footprints, and minimizing the need for external power sources.

- This project focuses on developing an automated solar-powered grass cutter that eliminates the need for human intervention.
- It utilizes solar energy to power both the robot and the cutting mechanism, making it an eco-friendly alternative to fuel-powered mowers.
- The system incorporates ultrasonic sensors for obstacle detection, ensuring safe operation.
- By using renewable solar energy, it reduces operational costs and environmental impact.
- The design aims to increase efficiency with dual blades for faster grass cutting and a self-recharging battery system.
- This innovation promotes sustainability, offering a cost-effective, green solution for lawn care.
- This project not only provides a cleaner and more sustainable alternative to conventional grass cutters but also demonstrates the potential of integrating renewable energy and robotics to address everyday challenges in a more efficient and environmentally responsible manner.

1.1 OBJECTIVE OF THE PROJECT:

➤ Primary Objectives

- **Design and Development of a Solar-Powered Robotic Grass Cutter:** The main objective is to design and create a robotic grass cutter that uses solar energy as its primary power source. The robot should be able to autonomously navigate and cut grass efficiently without human intervention.
- **Integration of Arduino for Automation:** The robot will be controlled by an Arduino microcontroller, which will manage the robot's movement, obstacle detection, and

cutting operations. The objective is to program the Arduino to enable autonomous decision-making, such as path finding and avoiding obstacles.

- **Optimization of Solar Energy Usage:** The solar-powered system should be designed to efficiently harness solar energy through solar panels to charge the robot's battery. The robot should be able to operate for extended periods using only solar power, optimizing energy consumption and minimizing reliance on external power sources.
- **Effective Grass Cutting Performance:** Ensure that the robotic system cuts the grass uniformly and effectively, with precision and consistent results across different lawn types. The robot should also be capable of navigating different terrains with minimal human intervention.

➤ Secondary Objectives

- **Improved User Experience:** Develop a user-friendly interface (if necessary) that allows users to monitor the robot's operation and make adjustments, such as controlling its start, stop, or scheduling functions.
- **Minimized Maintenance and Durability:** Design the robot to be durable and require minimal maintenance. This includes using weather-resistant materials and creating an efficient system that reduces the need for frequent repairs or replacements.
- **Safety Features:** Implement safety mechanisms to ensure the robot operates safely around humans, pets, and other obstacles. These features could include automatic shutdown or speed reduction when an obstacle is detected.
- **Cost-Effective Design:** Aim for a design that is cost-effective to produce and maintain, ensuring that the robot remains affordable for consumers while maintaining high performance.

1.2 Background

This Project is fully automated solar-powered grass cutting robot. Solar panel charges the battery, eliminating the need for external charging. Arduino microcontroller controls the robot and grass cutter motor functions. Ultrasonic sensor detects obstacles and ensures safety. The robot turns to avoid obstacles and resumes cutting when clear. Focus on automation and human-free operation of the grass cutting process. Designed to prevent damage to objects, humans, or animals. System integrates Arduino ATmega328p, ultrasonic sensors, and motor driver circuit for smooth operation.

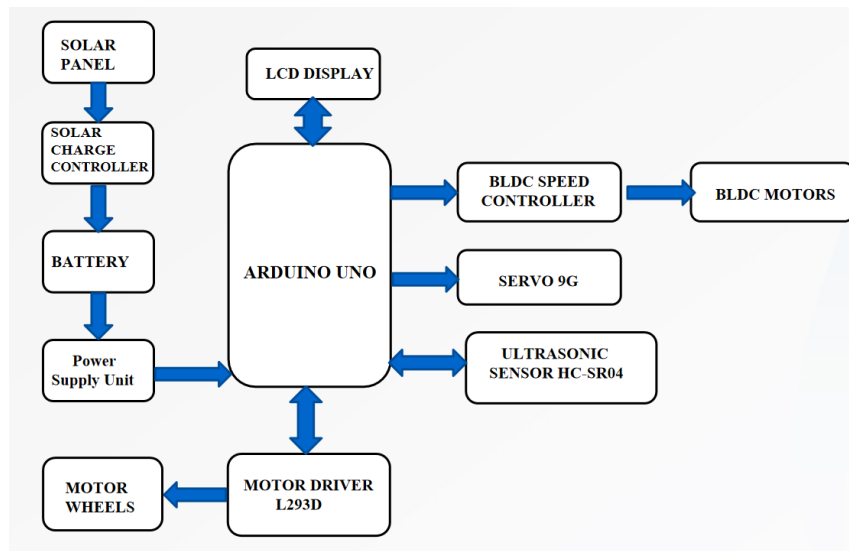


Fig. 1.1: Block diagram of solar powered robotic grass cutter

1. Solar Panel:

Solar panel uses light energy (photons) from sun to generate electricity through the photovoltaic effect. Electrical connections are made in series to achieve desired output voltage. It is also connected in parallel to achieve the rated current capacity of panel. Solar power generates electricity which is used to charge the batteries. The energy from sunlight is absorbed by the solar panel. The solar panel has an array of solar cells which convert the absorbed energy to electrical energy by the phenomenon of photovoltaic effect

2.Arduino UNO :The **Arduino Uno** is a versatile microcontroller board widely used in electronics projects, prototyping, and education. Powered by the ATmega328P microcontroller, it operates at 5V with a clock speed of 16 MHz. It features 14 digital I/O pins (6 with PWM), 6 analog input pins, and built-in communication interfaces like USB and ICSP headers. The Uno is powered via USB, external adapters, or batteries and includes flash memory (32 KB), SRAM (2 KB), and EEPROM (1 KB). It is programmed using the Arduino IDE, which supports a simplified C/C++ language, making it ideal for creating interactive devices and automating tasks.

3. Charge Controller:

The amount of solar power absorbed by panel is never constant as it varies due to the different factors such as sun position, cloudy weather, tilt angle etc. So, the electricity generated by solar panel is continuously varying in nature. This varying voltage effect requires a device

which can provide a constant Dc Output. Charge controller prevents over charging and protects against over voltage.

4. L293D Motor driver: The **L293D** is a popular motor driver IC used to control DC motors, stepper motors, and other inductive loads in robotics and automation projects. It can drive motors in both directions, allowing for forward and reverse control. The L293D can handle up to 600mA per channel and peaks up to 1.2A, with built-in diodes for back EMF protection. It operates at 4.5V to 36V, making it suitable for a wide range of motor control applications. It is typically controlled via logic signals from a microcontroller.

5. Ultrasonic Sensor HC-SR04 : The **HC-SR04** is a widely used ultrasonic sensor for measuring distances. It consists of a transmitter and receiver that emit high-frequency sound waves and measure the time it takes for the waves to bounce back from an object. By calculating the time delay, it determines the distance to the object. The sensor can measure distances between 2 cm and 4 meters with a precision of around 3mm. It is commonly used in robotics, object detection, and range-finding applications, interfacing easily with microcontrollers like Arduino.

1.4 Problem statement

1. The problem addressed by this project is the reliance on fuel-powered lawn movers, which contribute to environmental pollution, high operational costs, and noise pollution.
2. As fuel prices continue to rise and emissions from gas-powered machines increase, there is a need for an alternative solution.
3. This project aims to develop a solar-powered grass cutting machine that eliminates the need for fuel, reduces harmful emissions, offers a quieter operation, and provides a cost-effective and sustainable option for lawn care.

Chapter 2

LITERATURE SURVEY

1. **“Estimation for SOC of Electric Vehicle Lithium Battery Based on Artificial Immune Particle Filter”** Cheng He¹, Changchun Liu^{*2}, Yang Wu³, Tao Wu -2018-IEEE

In order to accurately predict the state of charge value of the electric vehicle, based on the characteristics of the collected data and combined with the empirical battery model, a battery state space model suitable for pure electric vehicles is established.

2. **“The Role of Battery Operated Auto-Rickshaw in the Transportation System of a City”** by SohelRana, Fahim Hossain, Shuvangkor Shusmoy Roy, and Suman Kumar Mitra-2013-Journal of Asian Electric Vehicles

The paper contains primary data based analysis on the role played by Battery Operated Auto-rickshaws in urban transportation system. Different aspects of battery operated auto-rickshaw, such as energy consumption rate, transport related emission attenuation rate, potentials of generating income and employment, trip characteristics, problems associated etc have been explored with a view to providing a generalized idea on the mode

3. **“E-Rickshaw Service in Barddhaman Town: Importance, Problems and Future Prospects”** by Arindam Roy, International Journal of Scientific and Research Publications, Volume 6, Issue 9, September 2016

The author describes that due to rapid rate of urban development along with increasing environmental problems, a strong need for eco-friendly transportation is needed. The study explains the importance of e-rickshaw service in the urban transport sphere of Barddhaman, west Bengal.

4. **“Comparison of Path Tracking and Torque-Vectoring Controllers for Autonomous Electric Vehicles”** Christoforos Chatzikomis, Aldo Sorniotti, IEEE Transactions on Intelligent Vehicles, Volume: 3 , Issue: 4 , Dec. 2018.

This paper analyzes the benefits of torque vectoring in an autonomous electric vehicle, either by integrating the torque-vectoring system into the path tracking controller, or through its separate implementation alongside the steering controller for path tracking. A

selection of path tracking controllers is compared in obstacle avoidance tests simulated with an experimentally validated vehicle dynamics model.

5. **“Electric Tricycle for Commercial Transportation”** Arinze David, Adisa Adelakun, Etukudor Christie, Ajayi Femi- International Conference on African Development Issues 2016

The paper analyses the design of an electric solar powered tricycle for use as a commercial means of transportation. The tricycle uses an electric brushless direct current motor connected to the rear wheels of the tricycle using the chain and sprocket mechanism. This motor is powered by direct current from the battery bank. The battery bank is charged via a solar PV system directly installed on the roof of the tricycle. This enables the panel to charge the battery bank while the tricycle is in motion.

6. **“Autonomy Analysis of a Solar Electric Tricycle”** Mohamed Dahbi, Said Doubabi, 3rd International Renewable and Sustainable Energy Conference (IRSEC), 2015.

Recent years, tricycles are increasing since, they can be used for tourist transportation as well as for transporting goods. In the light of this vision, the main idea of this paper is to study the performances of an electric tricycle equipped with solar panels. This study proposes an adaptation of the electric tricycles to become equipped with solar panels in order to increase batteries autonomy. Simulations of different topologies are presented along with calculation of solar panels power losses and batteries autonomy. A solution for increasing batteries autonomy is also discussed.

7. **“Design and Implementation of Ecofriendly vehicle and its Impact on Environment”** Prashant Kumar, Dr. Sanjay A. Deokar, Dr. Ajit M. Kate, 3rd International Conference on Communication and Electronics Systems (ICCES), 2018

Solar energy is greater source of non-renewable energy. We can harness the solar energy to run the vehicles. In this paper our main objective is to create a prototype of a solar based vehicle which is a tricycle. The solar vehicle can be a cost effective and environment friendly solution for daily usage of mode of transportation. The prototype consists of a charge controller, solar panel, DC hub motor and battery.

Chapter 3

METHODOLOGY

Grass cutter, which has been developed in this work, is a fully automated robotic vehicle without any human intervention, which uses solar energy to operate. The main advantage of this vehicle is, it detects obstacles and takes the deviation with the help of ultrasonic range sensors and microcontrollers. The working process of the grass cutter consists of a motor, which provides high rotational speed to the blades.

Due to the high speed of blades the grass gets trimmed by shearing action. In the current situation manually handled devices are used for trimming the grass, which desires an operator to move the vehicle yourself. By using Arduino uno based solar grass cutter the labor cost, maintenance cost can be eliminated to a certain extent compared to manually operated machine.

Based on a literature survey, it is found that, currently almost all the grass cutter devices work with conventional fuel and electricity, which causes pollution. In order to overwhelm these problems, a completely automated solar grass cutter has been established, which supports the green technology initiative, by dropping the pollution.

The major components of Arduino uno based solar grass cutter model are solar panel, batteries, microcontroller, sensors, motor driver, Bluetooth, DC motors and cutter. The abundant solar energy is collected, with the help of a solar panel which is used as a source of energy. The battery is used to store the electrical energy produced by the solar panel. The design contains a microcontroller, multiple sensors, and a solar charging system. Adding these elements together, we get our robotic lawn mower. The sensors are the eyes of our Robot. Initially, we had an idea what type of sensors we wanted to use. We used only an ultrasonic sensor to detect if the robot was heading into an object. Safety is the main concern when designing a robot with blades. We wanted our robot not to start operating if it was being held in the air by the user. Knowing that the user would be randomly holding the robot we needed a sensor to detect orientation. The power the system there are many options. With recharging batteries, there are various chemistries but we decided to go with the one that work best with solar charging. The **Arduino Uno-based solar grass cutter** is an innovative, fully automated robotic vehicle designed to trim grass without human intervention, powered by solar energy. This system uses an ultrasonic sensor to detect obstacles and adjust its path. A motor drives high-speed blades to shear the grass, eliminating the need for manual operation. The solar panel collects energy, which is stored in a battery for continuous operation. The

microcontroller, along with sensors, motor drivers, and Bluetooth, ensures smooth functioning. Unlike conventional fuel-powered or electric grass cutters, this solar-powered device supports green technology by reducing pollution and maintenance costs while offering a more sustainable solution.

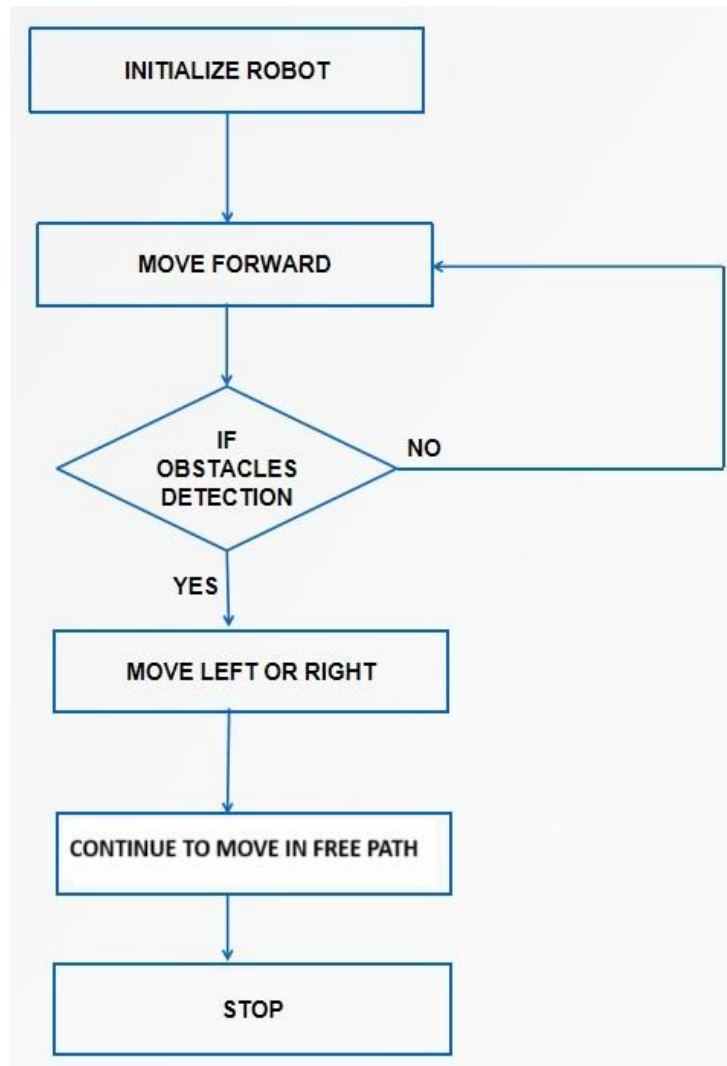


Fig. 3.1: Flow diagram of the solar powered robotic grass cutter using arduino

Chapter 4

HARDWARE COMPONENTS

A battery is a device that converts chemical energy directly into electrical energy. It consists of a number of voltaic cells; each voltaic cell consists of two half-cells connected in series by a conductive electrolyte containing cations and anions.

A battery fulfils the two main functions:

1. They are portable sources of electric power;
2. They are used to store the electrical energy.

In practice the words cell and battery mean closed electrochemical power sources. The primary cell or battery is a system whose useful life is ended once its reactants have been consumed by the discharge process. But the secondary battery is capable of being charged or recharged when its reactants have been used up. The spontaneous electrochemical reaction can be reversed by passing current through the cell in the opposite direction to that of cell discharge. It means the secondary battery might be considered as an electrochemical energy storage unit. Sometimes other terms are used to describe this type of system, e. g. accumulator, storage battery and so on

A cell may be characterised in terms as follows:

1. Available capacity;
2. Available energy;
3. The power it can deliver.

Currently most electric three-wheelers in India use lead acid batteries, which have a shorter life, duration, long charging time of 10 hours, and excess weight. Lead acid batteries typically have to be replaced every year and weigh around 120kg for an e-rickshaw application.

While the lithium ion battery costs around Rs 55,000-60,000 more than a conventional lead acid battery, the additional cost can be offset by the much longer battery life, higher confidence of financiers, service back-up and fast charging facility.

4.1 Design of Battery Ratings

Gross weight = 210 kg

Maximum velocity required = $V = 40\text{km/hr} = 11.11\text{m/s}$

Wheel size = 0.110 m

Area of vehicle $A = \text{length} * \text{breadth}$

$$= 2.625 * 1.3 = 3.41 \text{ m}^2$$

Linear distance travelled = $2\pi r = 0.6911 \text{ m}$

Speed in RPM = $40000 / (0.6911 * 60) = 965 \text{ rpm}$

Power = $[(\text{mass in kg}) * (g) * (V \text{ m/s}) * \text{rolling resistance}] + [\text{air density} * \text{co-efficient of drag} * \text{area of vehicle} * V^3]$

$$= [210 * 9.81 * 11.11 * 0.01] + [0.64 * 0.88 * 3.41 * 1371]$$

$$= 2889 \text{ watts} = 3000 \text{ W}$$

Battery calculations

Distance per charge = 60 km at 40km/hr velocity

Power output/km = $3000 / 40 = 75 \text{ Wh/km}$

Let the voltage of motor be 48V

AH used per km = $75 / 48 = 1.56 \text{ AH/km}$

Total AH required = $1.56 * 60 * 1.25$

$$= 117 \text{ AH} = 120 \text{ AH battery}$$

From the above calculations we can choose 48V, 120 AH battery for the proposed work

4.2 Lithium ion verses lead acid batteries

4.2.1 Lithium ion batteries

- In a lithium-ion battery, high energy density leads to low weight which translates into more kilometres between charges. While a Li-Ion battery weighs 35kg, a 48 V-80 Ah lead acid battery weighs 120kg.
- While a Li -Ion unit can be charged in only 1.5 to 3 hours, lead acid battery charging time is 8-10 hours or more. Li-Ion cells have a life of 1,500 cycles (NMC) and 3,000 cycles (LeFePo4) – a life up to 5 years. In comparison, lead acid batteries have a typical life of 360 cycles.
- Li-Ion batteries have a low self-discharge which *means* if the vehicle is not used for some time, its charge status remains the same.- They are also low maintenance. Li-Ion cells, unlike lead acid or nickel cadmium, do not require maintenance to ensure their performance.

4.2.2 Lead acid battery

Despite having a very low energy-to-weight ratio and a low energy-to-volume ratio, its ability to supply high surge currents means that the cells have a relatively large power-to-weight ratio. These features, along with their low cost, make them attractive for use in motor vehicles to provide the high current required by automobile starter motors.

The storage battery or secondary battery is such a battery where electrical energy can be stored as chemical energy and this chemical energy is then converted to electrical energy as and when required. The conversion of electrical energy into chemical energy by applying external electrical source is known as charging of battery. The conversion of chemical energy into electrical energy for supplying the external load is known as discharging of secondary battery. During charging of battery, current is passed through it which causes some chemical changes inside the battery. These chemical changes absorb the energy during their formation. When the battery is connected to the external load, the chemical changes take place in reverse direction, during which the absorbed energy is released as electrical energy and supplied to the load. Now we will try to understand the principle working of lead acid battery and for that we will first discuss about lead acid battery which is very commonly used as storage battery or secondary battery.

Table 4.1: Lead acid battery specifications

Usage/Application	E Rickshaw
Battery Type	Sealed Lead Acid Battery
Weight	35 kg Approx
Application	E Rickshaw
Charging Time	7-8 Hours
Volt	12 Volt
Minimum Order Quantity	20 Unit

Definition: The battery which uses sponge lead and lead peroxide for the conversion of the chemical energy into electrical power, such type of battery is called a lead acid battery. The

lead acid battery is most commonly used in the power stations and substations because it has higher cell voltage and lower cost.

Construction of Lead Acid Battery

The various parts of the lead acid battery are shown below. The container and the plates are the main part of the lead acid battery. The container stores chemical energy which is converted into electrical energy by the help of the plates.

1. Container – The container of the lead acid battery is made of glass, lead lined wood, ebonite, the hard rubber of bituminous compound, ceramic materials or moulded plastics and are seated at the top to avoid the discharge of electrolyte. At the bottom of the container, there are four ribs, on two of them rest the positive plate and the others support the negative plates. The prism serves as the support for the plates and at the same time protect them from a short-circuit. The material of which the battery containers are made should be resistant to sulphuric acid, should not deform or porous, or contain impurities which damage the electrolyte.

2. Plate – The plate of the lead-acid cell is of diverse design and they all consist some form of a grid which is made up of lead and the active material. The grid is essential for conducting the electric current and for distributing the current equally on the active material. If the current is not uniformly distributed, then the active material will loosen and fall out.

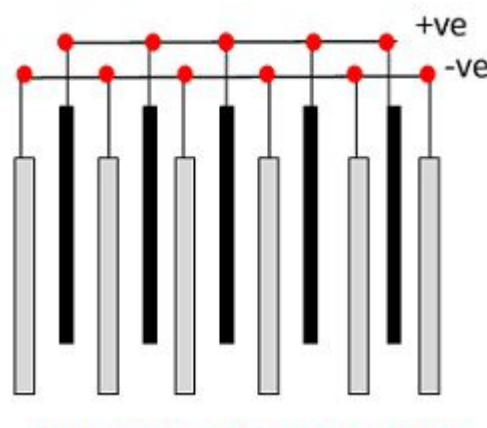


Fig 4.1: arrangement of plates in lead acid battery

The grids are made up of an alloy of lead and antimony. These are usually made with the transverse rib that crosses the places at a right angle or diagonally. The grid for the positive and negative plates are of the same design, but the grids for the negative plates are made lighter because they are not as essential for the uniform conduction of the current.

The plates of the battery are of two types. They are the formed plates or plante plates and pasted or Faure plates.

Plante's plates are used largely for stationary batteries as these are heavier in weight and more costly than the pasted plates. But the plates are more durable and less liable to lose active material by rapid charging and discharging. The plante's plate has low capacity weight-ratio. Faure process is much suitable for manufacturing of negative plates rather than positive plates. The negative active material is quite tough, and it undergoes a comparatively low change from charging and discharging.

3. Active Material – The material in a cell which takes active participation in a chemical reaction (absorption or evolution of electrical energy) during charging or discharging is called the active material of the cell. The active elements of the lead acid are

1. **Lead peroxide (PbO_2)** – It forms the positive active material. The PbO_2 are dark chocolate broom in colour.
2. **Sponge lead** – Its form the negative active material. It is grey in colour.
3. **Dilute Sulphuric Acid (H_2SO_4)** – It is used as an electrolyte. It contains 31% of sulphuric acid.

The lead peroxide and sponge lead, which form the negative and positive active materials have the little mechanical strength and therefore can be used alone.

4. Separators – The separators are thin sheets of non-conducting material made up of chemically treated Leadwood, porous rubbers, or mat of glass fibre and are placed between the positive and negative to insulate them from each other. Separators are grooved vertically on one side and are smooth on the other side.

5. Battery Terminals – A battery has two terminals the positive and the negative. The positive terminal with a diameter of 17.5 mm at the top is slightly larger than the negative terminal which is 16 mm in diameter.

Working Principle of Lead Acid Battery

When the sulphuric acid dissolves, its molecules break up into positive hydrogen ions (2H^+) and sulphate negative ions (SO_4^-) and move freely. If the two electrodes are immersed in solutions and connected to DC supply then the hydrogen ions being positively charged and moved towards the electrodes and connected to the negative terminal of the supply. The SO_4^- ions being negatively charged moved towards the electrodes connected to the positive terminal of the supply main (i.e., anode).

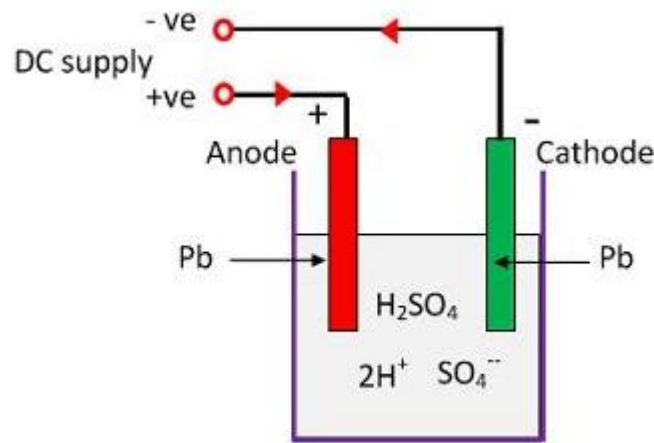


Fig 4.2: Charging of lead acid battery

Each hydrogen ion takes one electron from the cathode, and each sulphates ions takes the two negative ions from the anodes and react with water and form sulphuric and hydrogen acid.

The oxygen, which produced from the above equation react with lead oxide and form lead peroxide (PbO_2 .) Thus, during charging the lead cathode remain as lead, but lead anode gets converted into lead peroxide, chocolate in colour.

If the DC source of supply is disconnected and if the voltmeter connects between the electrodes, it will show the potential difference between them. If wire connects the electrodes, then current will flow from the positive plate to the negative plate through external circuit i.e. the cell is capable of supplying electrical energy.

Chemical Action during Discharging

When the cell is full discharge, then the anode is of lead peroxide (PbO_2) and a cathode is of metallic sponge lead (Pb). When the electrodes are connected through a resistance, the cell discharge and electrons flow in a direction opposite to that during charging.

The hydrogen ions move to the anode and reaching the anodes receive one electron from the anode and become hydrogen atom. The hydrogen atom comes in contacts with a PbO_2 , so it attacks and forms lead sulphate ($PbSO_4$), whitish in colour and water according to the chemical equation.

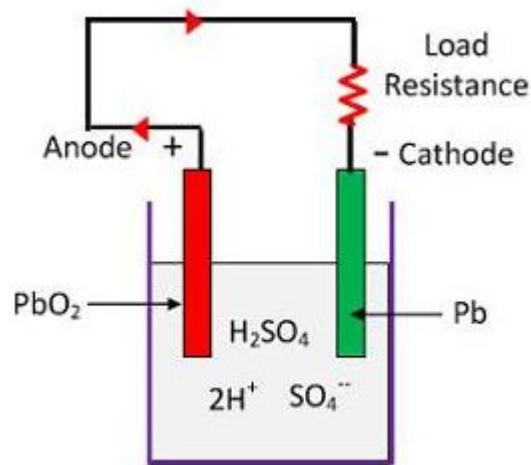


Fig 4.3: Discharging of lead acid battery

The each sulphate ion (SO_4^-) moves towards the cathode and reaching there gives up two electrons becomes radical $\text{SO}_4\cdot$, attack the metallic lead cathode and form lead sulphate whitish in colour according to the chemical equation.

Chemical Action during Recharging:

For recharging, the anode and cathode are connected to the positive and the negative terminal of the DC supply mains. The molecules of the sulphuric acid break up into ions of 2H^+ and SO_4^- . The hydrogen ions being positively charged moved towards the cathodes and receive two electrons from there and form a hydrogen atom. The hydrogen atom reacts with lead sulphate cathode forming lead and sulphuric acid according to the chemical equation.

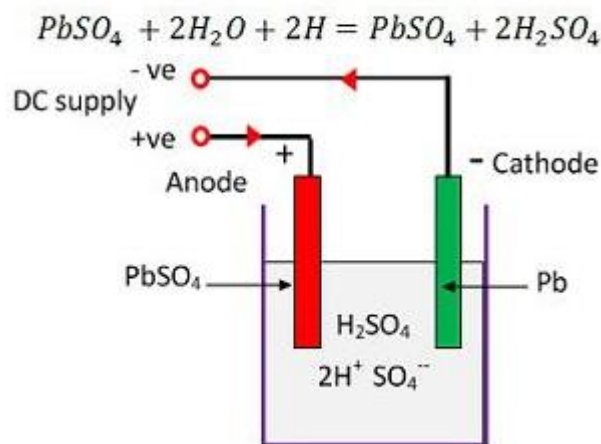
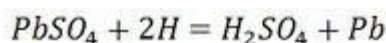


Fig. 4.4: Recharging of lead acid battery

Radical $\text{SO}_4\cdot$, react with the lead sulphate anode and form leads peroxide and lead sulphuric acid according to the chemical equation.



The charging and discharging are represented by a single reversible equation given below.

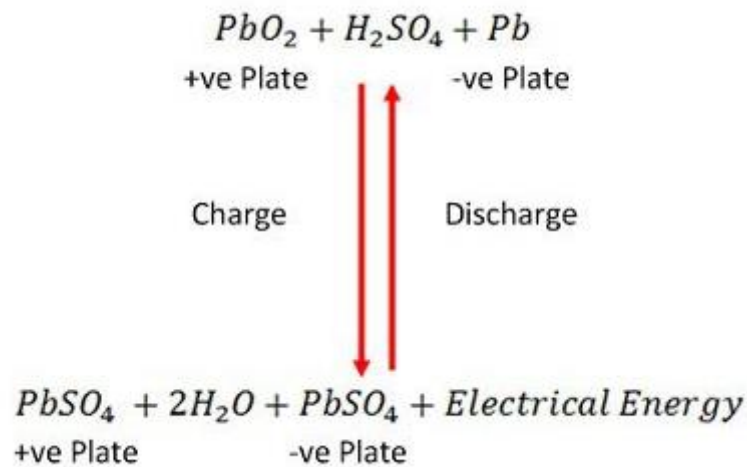


Fig. 4.5: Charging and discharging process

The equation should read downward for discharge and upward for recharge.

4.3 Maintenance Precaution

Ammonia can neutralize spilled battery acid. Surplus ammonia and water evaporate, leaving an ammonium sulphate residue. Sodium bicarbonate (baking soda) is also commonly used for this purpose.

4.3.1 Lithium ion battery

Like any other battery, a rechargeable lithium-ion battery is made of one or more power-generating compartments called **cells**. Each cell has essentially three components: a **positive electrode** (connected to the battery's positive or + terminal), a **negative electrode** (connected to the negative or – terminal), and a chemical called an **electrolyte** in between them. The positive electrode is typically made from a chemical compound called lithium-cobalt oxide (LiCoO_2) or, in newer batteries, from lithium iron phosphate (LiFePO_4). The negative electrode is generally made from carbon (graphite) and the electrolyte varies from one type of battery to another—but isn't too important in understanding the basic idea of how the battery works.

All lithium-ion batteries work in broadly the same way. When the battery is charging up, the lithium-cobalt oxide, positive electrode gives up some of its lithium ions, which move through the electrolyte to the negative, graphite electrode and remain there. The battery takes in and stores energy during this process. When the battery is discharging, the lithium ions move back

across the electrolyte to the positive electrode, producing the energy that powers the battery. In both cases, electrons flow in the opposite direction to the ions around the outer circuit. Electrons do not flow through the electrolyte: it's effectively an insulating barrier, so far as electrons are concerned

The movement of ions (through the electrolyte) and electrons (around the external circuit, in the opposite direction) are interconnected processes, and if either stops so does the other. If ions stop moving through the electrolyte because the battery completely discharges, electrons can't move through the outer circuit either—so you lose your power. Similarly, if you switch off whatever the battery is powering, the flow of electrons stops and so does the flow of ions. The battery essentially stops discharging at a high rate (but it does keep on discharging, at a very slow rate, even with the appliance disconnected).

Unlike simpler batteries, lithium-ion ones have built in electronic controllers that regulate how they charge and discharge. They prevent the overcharging and overheating that can cause lithium-ion batteries to explode in some circumstances.

As their name suggests, lithium-ion batteries are all about the movement of lithium ions: the ions move one way when the battery charges (when it's absorbing power); they move the opposite way when the battery discharges (when it's supplying power):

1. During charging, lithium ions (yellow circles) flow from the positive electrode (red) to the negative electrode (blue) through the electrolyte (gray). Electrons also flow from the positive electrode to the negative electrode, but take the longer path around the outer circuit. The electrons and ions combine at the negative electrode and deposit lithium there.
2. When no more ions will flow, the battery is fully charged and ready to use.
3. During discharging, the ions flow back through the electrolyte from the negative electrode to the positive electrode. Electrons flow from the negative electrode to the positive electrode through the outer circuit, powering your laptop. When the ions and electrons combine at the positive electrode, lithium is deposited there.
4. When all the ions have moved back, the battery is fully discharged and needs charging up again.

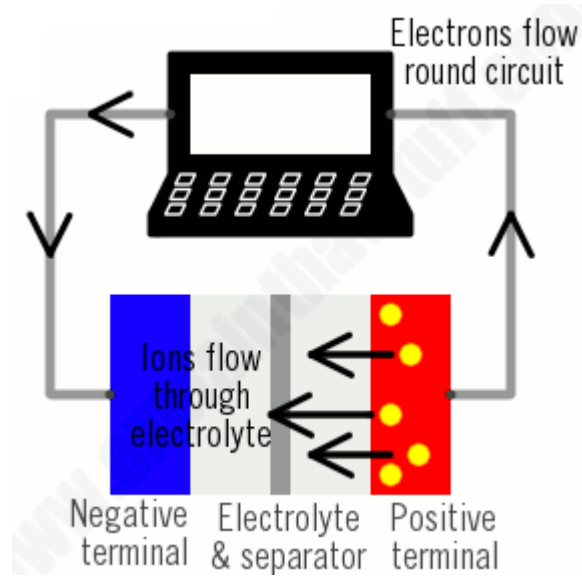


Fig 4.6: Movement of ions in lithium ion battery



Fig 4.7: Lithium ion battery

4.4 Charge controller or Battery management system

A solar charge controller is fundamentally a voltage or current controller to charge the battery and keep electric cells from overcharging. It directs the voltage and current hailing from the solar panels setting off to the electric cell. Generally, 12V boards/panels put out in the ballpark of 16 to 20V, so if there is no regulation the electric cells will damage from overcharging. Generally, electric storage devices require around 14 to 14.5V to get completely charged. The solar charge controllers are available in all features, costs, and sizes. The range of charge controllers is from 4.5A and up to 60 to 80A.

Features of Solar Charge Controller:

- Protects the battery (12V) from overcharging

- Reduces system maintenance and increases battery lifetime
- Auto charged indication
- Reliability is high
- 10amp to 40amp of charging current
- Monitors the reverse current flow

The function of the Solar Charge Controller:

The most essential charge controller basically controls the device voltage and opens the circuit, halting the charging, when the battery voltage ascends to a certain level. More charge controllers utilized a mechanical relay to open or shut the circuit, halting or beginning power heading off to the electric storage devices

Generally, solar power systems utilize 12V of batteries. Solar panels can convey much more voltage than is obliged to charge the battery. The charge voltage could be kept at the best level while the time needed to completely charge the electric storage devices is lessened. This permits the solar systems to work optimally constantly. By running higher voltage in the wires from the solar panels to the charge controller, power dissipation in the wires is diminished fundamentally.

The solar charge controllers can also control the reverse power flow. The charge controllers can distinguish when no power is originating from the solar panels and open the circuit separating the solar panels from the battery devices and halting the reverse current flow.

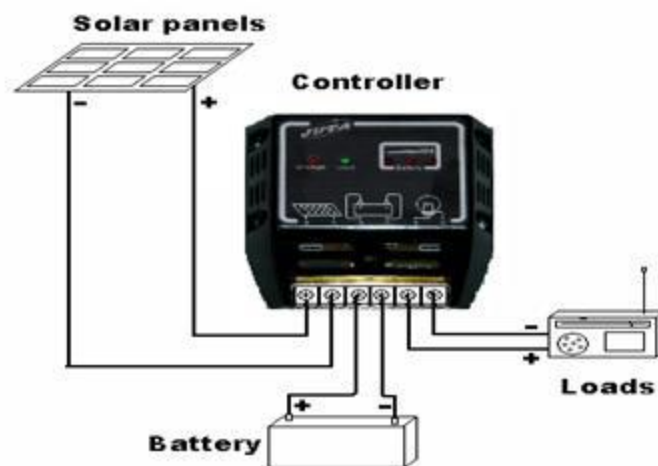


Fig. 4.8: charge controller

Battery management system (BMS):

A battery management system manages the rechargeable battery by providing protection to the battery. It also monitors the battery state, calculates and reports the secondary data etc. the BMS is a very important block in an EV. It is required to keep battery stable. The basic block diagram of BMS is shown below.

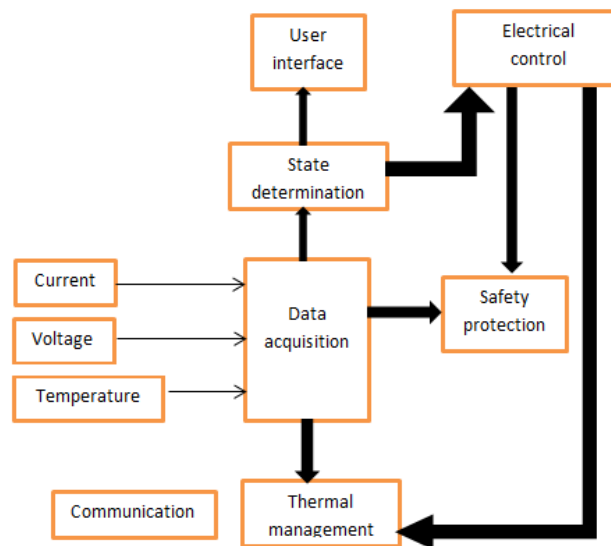


Fig.4.8: Basic block diagram of BMS

The BMS is mainly used to monitor the battery. The battery monitoring requires three important parameters such as the temperature, current, and voltage of each cell of battery. The other use of BMS is estimate the state of the Battery. This is done by measurement of state of charge (SOC) and state of health (SOH). SOC can be found using the voltage and current of cell, charging & discharging profile. SOH can be found from number of cycle of charge and battery performance.

4.5 Other components

The other components that are required for E-Rickshaw are,

- Brushless DC Motor 24V ,1A ,12V Operation
- Solar panel 150W,12V
- Speed controller
- Motor Drive
- Horn
- Display
- Head Lamp

- Forward and Backward switch
- Single pole triple switch
- Indicators
- Voltage and current display

Other specifications:

Charge Voltage	220 V
Top speed	30km/h
Continued trip mileage	35-40 km
Dimension (l*b*h)	2635mm *1300mm *1700mm
Net weight	150kg
Max. load capacity	Up to 300kg

Specifications and number of components used:

Component name	Specification	Number
Lead acid battery	12V ,15AH	1
Lithium ion battery	12V ,20AH	1
BLDC Motor	24V,1A,12V Operation	2
Solar panel	150W, 12V	1
Charge controller	5A, 12V	2

Chapter 5

WORKING OF SOLAR POWERED ROBOTIC GRASS CUTTER

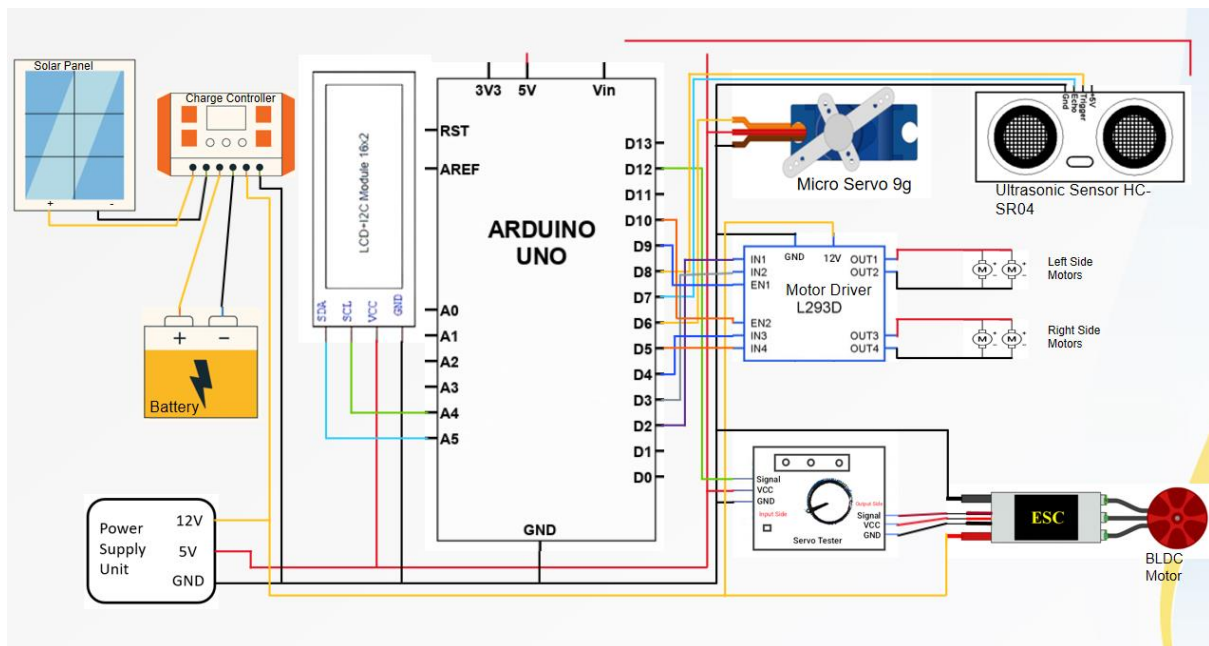


Fig.5.1: Circuit diagram of the solar powered robotic grass cutter using arduino

In the proposed system two BLDC motors are used with bolt drive and axil drive, and two batteries.

- The motors are powered by either lithium ion battery or lead acid battery.
- The batteries are charged by two different ways i.e lithium ion battery is charged by 230V AC supply where as lead acid battery is charged by a solar panel and a wind .
- Lead acid battery is only charged during day time and is used for emergency purpose.
- Lithium ion battery is charged by 230V AC supply which is reduced to 14.5V by rectifier.
- Each batteryhas a separate charge controller which is used to manage the current and to supply a constant 12V supply to the battery.
- The charge controller is directly connected to on/off switch(single pole triple throw) and then connected to forward and backward switch which is connected to motor drive and motors.
- All the other components like side indicator, horn, voltage and ampere display, speed controller, head lamp are connected independently

5.1 Components used for the project

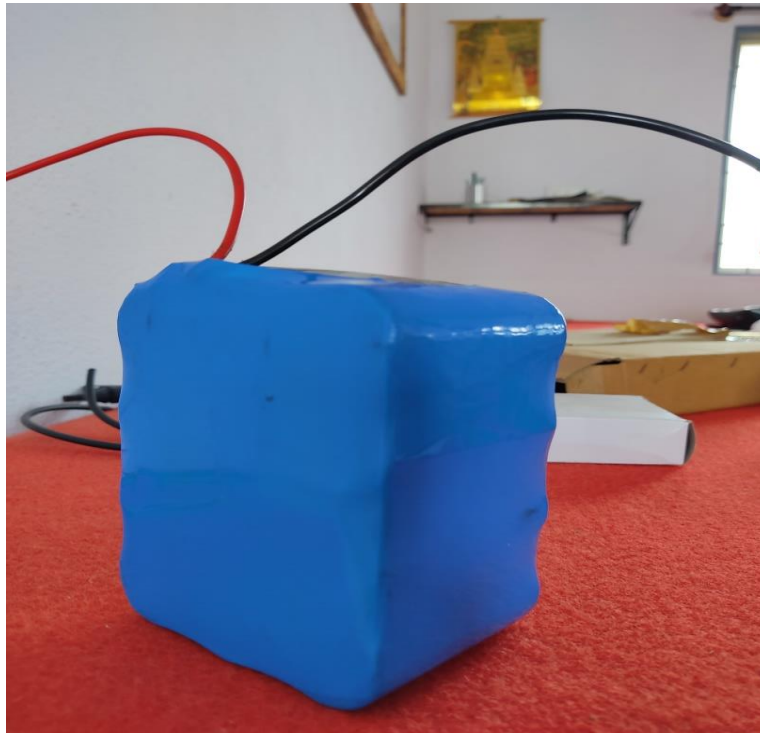


Fig 5.2:Lithium Ion battery



Fig 5.3: lead acid battery



Fig 5.4: Battery charge controller or battery management system



Fig 5.5: Switch used to switch over between batteries

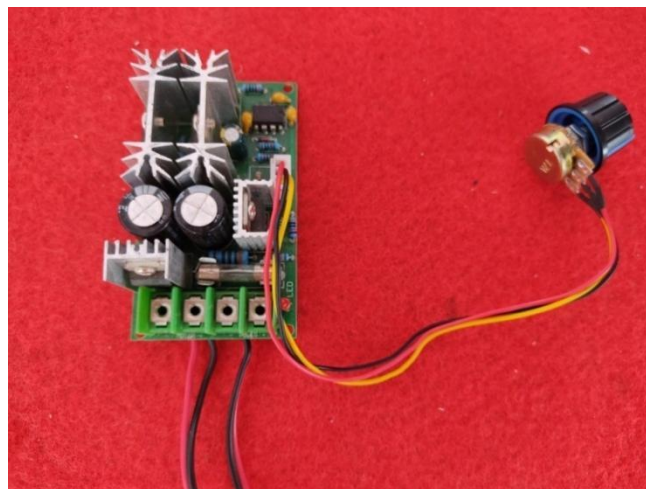


Fig 5.6: Motor driver

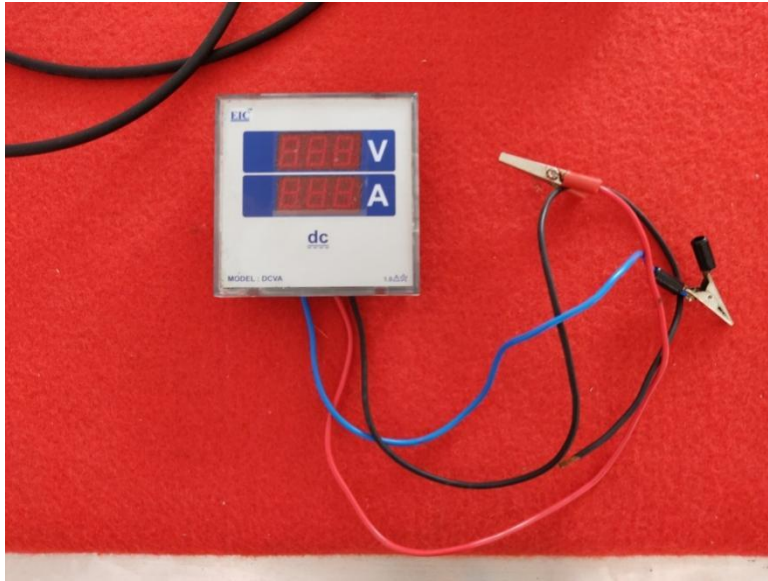


Fig 5.7: Voltage and ampere reading



Fig 5.8: Solar panel

Chapter 6

SOURCE CODE

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Motor control pins
#define LEFT_MOTOR_IN1 2
#define LEFT_MOTOR_IN2 3
#define RIGHT_MOTOR_IN3 4
#define RIGHT_MOTOR_IN4 5

#define LEFT_ENABLE 9 // PWM pin for left motor speed control
#define RIGHT_ENABLE 10 // PWM pin for right motor speed control

// Ultrasonic sensor pins
#define TRIG_PIN 8
#define ECHO_PIN 7

// Servo motor pin
#define SERVO_PIN 6

// Grass cutter motor relay pin
#define RELAY_PIN 12

Servo servo; // Create a servo object
LiquidCrystal_I2C lcd(0x27, 16, 2); // LCD with I2C

void setup() {
  pinMode(LEFT_MOTOR_IN1, OUTPUT);
  pinMode(LEFT_MOTOR_IN2, OUTPUT);
  pinMode(RIGHT_MOTOR_IN3, OUTPUT);
```

```
pinMode(RIGHT_MOTOR_IN4, OUTPUT);
pinMode(LEFT_ENABLE, OUTPUT);
pinMode(RIGHT_ENABLE, OUTPUT);
pinMode(TRIG_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);
pinMode(RELAY_PIN, OUTPUT);

servo.attach(SERVO_PIN);
servo.write(90); // Move servo to center position

lcd.begin(16, 2);
lcd.clear();
lcd.print("Robot Ready!");
Serial.begin(9600);
}

void loop() {
  int distance = measureDistance();

  if (distance > 50) {
    moveForward();
    lcd.setCursor(0, 1);
    lcd.print("Moving Forward");
    digitalWrite(RELAY_PIN, LOW); // Grass cutter ON
  } else {
    stopMotors();
    lcd.setCursor(0, 1);
    lcd.print("Obstacle Detected");
    digitalWrite(RELAY_PIN, HIGH); // Grass cutter OFF
    scanAndTurn();
  }
  delay(100);
}
```

```
int measureDistance() {  
    digitalWrite(TRIG_PIN, LOW);  
    delayMicroseconds(2);  
    digitalWrite(TRIG_PIN, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(TRIG_PIN, LOW);  
    long duration = pulseIn(ECHO_PIN, HIGH);  
    return duration * 0.034 / 2; // Convert to cm  
}
```

```
void moveForward() {  
    digitalWrite(LEFT_MOTOR_IN1, HIGH);  
    digitalWrite(LEFT_MOTOR_IN2, LOW);  
    digitalWrite(RIGHT_MOTOR_IN3, HIGH);  
    digitalWrite(RIGHT_MOTOR_IN4, LOW);  
    analogWrite(LEFT_ENABLE, 255);  
    analogWrite(RIGHT_ENABLE, 255);  
}
```

```
void moveBackward() {  
    lcd.setCursor(0, 1);  
    lcd.print("Moving Backward");  
    digitalWrite(LEFT_MOTOR_IN1, LOW);  
    digitalWrite(LEFT_MOTOR_IN2, HIGH);  
    digitalWrite(RIGHT_MOTOR_IN3, LOW);  
    digitalWrite(RIGHT_MOTOR_IN4, HIGH);  
    analogWrite(LEFT_ENABLE, 200);  
    analogWrite(RIGHT_ENABLE, 200);  
    delay(1000);  
    stopMotors();  
}
```

```
void stopMotors() {  
    digitalWrite(LEFT_MOTOR_IN1, LOW);
```

```
digitalWrite(LEFT_MOTOR_IN2, LOW);
digitalWrite(RIGHT_MOTOR_IN3, LOW);
digitalWrite(RIGHT_MOTOR_IN4, LOW);
}

void turnLeft() {
  lcd.setCursor(0, 1);
  lcd.print("Turning Left...");
  digitalWrite(LEFT_MOTOR_IN1, LOW);
  digitalWrite(LEFT_MOTOR_IN2, HIGH);
  digitalWrite(RIGHT_MOTOR_IN3, HIGH);
  digitalWrite(RIGHT_MOTOR_IN4, LOW);
  analogWrite(LEFT_ENABLE, 255);
  analogWrite(RIGHT_ENABLE, 255);
  delay(700); // Adjust duration for a 70-90 degree turn
}

void turnRight() {
  lcd.setCursor(0, 1);
  lcd.print("Turning Right...");
  digitalWrite(LEFT_MOTOR_IN1, HIGH);
  digitalWrite(LEFT_MOTOR_IN2, LOW);
  digitalWrite(RIGHT_MOTOR_IN3, LOW);
  digitalWrite(RIGHT_MOTOR_IN4, HIGH);
  analogWrite(LEFT_ENABLE, 255);
  analogWrite(RIGHT_ENABLE, 255);
  delay(700); // Adjust duration for a 70-90 degree turn
}

void scanAndTurn() {
  int leftDistance = 0;
  int rightDistance = 0;

  // Scan left
```

```
servo.write(0);
delay(500);
leftDistance = measureDistance();
Serial.print("Left Distance: ");
Serial.println(leftDistance);

// Scan right
servo.write(180);
delay(500);
rightDistance = measureDistance();
Serial.print("Right Distance: ");
Serial.println(rightDistance);

servo.write(90);
delay(500);

// Logic for turning based on obstacle detection
if (leftDistance < 50 && rightDistance >= 50) {
    turnLeft(); // Turn left if obstacle is on the left side and right side is clear
} else if (rightDistance < 50 && leftDistance >= 50) {
    turnRight(); // Turn right if obstacle is on the right side and left side is clear
} else {
    stopMotors(); // Stop if both sides are blocked
}

delay(1000); // Allow time for movement
stopMotors();
}
```

Chapter 7

ADVANTAGES & DISADVANTAGES OF E-RICKSHAW

7.1 Advantages of Grass Cutter :

- **Environmentally Friendly:** It uses solar energy, reducing reliance on fossil fuels and electricity, thereby lowering carbon emissions and supporting green energy initiatives.
- **Cost-Effective:** The use of solar power eliminates fuel and electricity costs, and the autonomous nature reduces the need for labor, making it more economical in the long term.
- **Noise Reduction:** Unlike gas-powered mowers, solar-powered robots produce significantly less noise, reducing noise pollution in residential areas.
- **Autonomous Operation:** The robotic grass cutter operates without human intervention, making it convenient and saving time for users.
- **Low Maintenance:** With fewer moving parts and no fuel-related components, the solar-powered mower requires less maintenance compared to traditional lawnmowers.
- **Safety:** Features like obstacle detection sensors and automatic stop mechanisms ensure safer operation, especially in unpredictable environments.
- **Sustainability:** Solar energy is renewable, ensuring the grass cutter operates without depleting non-renewable resources, contributing to long-term sustainability.

7.2 Limitations of Grass Cutter:

- **Limited Power in Cloudy Weather:** Solar energy generation depends on sunlight, so performance may be reduced on cloudy or rainy days, potentially requiring longer charging times.
- **Battery Life and Charging:** The efficiency of the battery is limited by solar charging, and if the battery runs out, it may require manual recharging, impacting the mower's autonomy.
- **High Initial Cost:** The upfront cost of the solar-powered robotic mower, including the solar panel and battery system, can be higher compared to traditional gas-powered or electric mowers.
- **Limited Coverage Area:** The effectiveness of the mower may be restricted by the size of the solar panel and the battery's charge capacity, limiting the area it can cover before needing to recharge.

7.3 Scope

By 2030 we are expecting only electric vehicles on the roads, and if we can convert the existing conventional vehicles to electric vehicles that will be cost efficient. Furthermore the storage capacity and charging time can also be improved by the entire reliability and efficiency of the system

Chapter 8

CONCLUSION

In conclusion, the Solar-Powered Robotic Grass Cutter represents a significant leap forward in lawn maintenance technology. By harnessing the power of solar energy and advanced robotics, this innovative system offers a sustainable, efficient, and cost-effective solution for lawn care. The Solar-Powered Robotic Grass Cutter is poised to transform the lawn care industry and contribute to a more environmentally friendly future. The Solar-Powered Robotic Grass Cutter is well-positioned to revolutionize the lawn care industry and contribute to a more sustainable future. This reduces noise pollution, and operational costs. The **Solar-Powered Robotic Grass Cutter** offers a groundbreaking advancement in lawn maintenance, combining the efficiency of robotics with the sustainability of solar energy. This innovative system reduces reliance on conventional fuel and electricity, cutting operational costs and minimizing environmental impact. By using solar power, it reduces carbon emissions and lowers noise pollution, making it an eco-friendly alternative to traditional lawnmowers. With its autonomous operation, it eliminates the need for manual labor and enhances convenience for homeowners. As the lawn care industry moves towards more sustainable practices, this robotic grass cutter is well-equipped to drive change and contribute to a greener future.

Chapter 9

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Chapter 10

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